



TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.111

Safeguarding RDK warning threshold air pressure

Test Carrier:	Roller Drum Test Rig
Testing Facility:	Test Rig of Tire Manufacturer
Status (Porsche):	25.11.2019
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(see target data sheet / "Zieldatenblatt")	

1 Purpose

The purpose of the test procedure is to ensure the RDK warning threshold air pressure of all tires for the respective vehicle and engine variants according to the valid target data sheet.

2 Field of Application

This test method specifies a procedure for testing the high-speed endurance of tires on a chassis dynamometer.

The test procedure is valid for:

- Test Tire Trials
- Approval Tests
- Series Monitoring

3 Definition of Terms

V_{max}:

Typed maximum vehicle speed (see target data sheet)

RDK warning threshold air pressure:

Air pressure according to target data sheet - 0.3 bar (unless another value is noted on the target data sheet)

Worst Case Condition High-Speed Test:

The most unfavorable load combination of speed, camber, air pressure and axle load selected by the tire manufacturer from the high-speed test program parameter table. The comfort air pressures must also be taken into account.

4 Responsibility

Test Tire Trial:

The inspection for the test tire delivery is performed and documented by the tire manufacturer (self-certification). One tire per specification shall be tested according to the "worst case" conditions selected by the tire manufacturer. The results shall be made available to the PAG immediately.

Approval Tests:

The test for approval is performed and documented by the tire manufacturer (self-certification). The results are made available to the PAG immediately. Two tires per specification shall be tested according to the "worst case" conditions selected by the tire manufacturer. The results will be made available to the PAG immediately.

Series Monitoring:

The monitoring of the series is carried out and documented by the tire manufacturer (self-certification). The test intervals and the number of test tires are at the discretion of the tire manufacturer. However, at least once a year two tires should be tested consecutively according to the "worst case" conditions selected by the tire manufacturer and the result should be reported to the PAG. In case of non-compliance, the PAG must be informed immediately.

Retesting:

Retesting of approval/production tires can be arranged by the PAG, Tire Department, at an independent institute.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

Measured Variable	Measurement Tolerance	Adjustment & Control Tolerance
Drum Speed	$\pm 1\%$ of measurand	± 1 km/h
Wheel Load	$\pm 1\%$ of measurand	± 50 N
Tire Pressure	± 0.1 bar	± 0.1 bar
Ambient Temperature	$\pm 0.5^\circ\text{C}$	-
Tire Surface Temp.	$\pm 0.5^\circ\text{C}$	-
Time	± 1 s	-

The tire pressure is measured at the beginning of the measurement. The tire surface temperature, if not continuously measurable, is measured at the beginning and if required (e.g. for the release test) at the end or up to three minutes after the end of the measurement. The ambient temperature is measured continuously.

5.1.2 Test Rig

- External Drum Test Rig with diameter 1.7m, 2.0m or 3.0m
- Datum diameter with 2.0m (different load reductions depending on diameter, see test parameters)
- Drum surface: steel, smooth
- Drum driven, speed controlled
- Wheel receiver with camber adjustment possibility up to $\pm 4^\circ$

5.1.3 Data Processing

N/A

5.2 Test Conditions

- Ambient temperature must be between 20°C and 30°C
- Reference temperature for air pressure setting is 20°C
- The tire manufacturer has to ensure all vehicle and air pressure variants by his tests on the basis of the valid target data sheet (parameter table for the high-speed test program)

5.3 Test Carrier / Test Set-Up

- The test load corresponds to the wheel load selected according to the worst-case condition in the target data sheet (parameter table for the high-speed test program) multiplied by:
 - 0.9, if drum diameter = 1.7m or 2.0m
 - 0.93, if drum diameter = 3.0m
- Setting the RDK warning threshold air pressure in an uncontrolled manner
- Camber angle depending on the respective vehicle value (see target data sheet)
- Skew angle = 0
- Rim width, diameter and contour at the rim seat of the test rim must correspond to the respective original Porsche wheel (see target data sheet)
- Additional drillings e.g. for internal temperature measurements are permissible

5.4 Test Procedure

The mounted wheel is stored under test room temperature for at least three hours before testing. Before the test, the RDK warning threshold air pressure is set. The drum speed is increased gradually according to the respective program in the following table. The acceleration phases of the drum should not exceed 30 seconds (one minute at the beginning).

Test Stages	Runtime (min)	Drum speed (km/h)
1	5	Vmax - 90
2	5	Vmax - 40
3	10	Vmax - 30
4	10	Vmax - 20
5	10	Vmax - 10
6	20	Vmax

For the high-speed tests, at least two tires per specification must be tested consecutively for all scenarios shown in the target data sheet. The test must not be interrupted, but runs until the runtime specification is met (target: 60 minutes)

6 Documentation

6.1 Data Analysis

The following data must be listed in the results sheet:

- Test date and time
- Tire size
- Tire make, designation and specification number
- DOT number
- Tire condition (new, used, tread depth)

- Designation of the test program
- Designation of the selected vehicle, loading and air pressure scenario, or alternatively a selected combination of wheel load, camber air pressure and speed as a "worst case" condition
- Documentation of the target data sheet used for testing (date/change status)
- Test parameters (load, air pressure, wheel dimensions, camber)
- Speed level at 1st damage occurrence
- Runtime at 1st damage occurrence
- Speed level at test termination
- Runtime at test termination
- Cause of damage, damage pattern (attach at least on photo)
- Ambient temperature per speed level and at the end of the test
- Tire surface temperature at the beginning and at the end of the test
- Tire pressure at the beginning and at the end of the test
- Name of the tester
- Test site, test rig

6.2 Presentation of the results

The presentation of the results for test tires as well as for the approval test 2U or BMG approval is to be executed in the form of the test catalog, data sheet Safeguarding RDK warning threshold air pressure F.360.111 for tire releases (release tests) of the PAG.

6.3 Requirements

Requirements for test tires, approval tests and series monitoring:

The test is considered to have been passed if test stages 1 to 6 (equivalent to 60 min runtime) are passed without damage.

7 Applicable Documents

Porsche AG, data sheet: "Safeguarding RDK warning threshold air pressure F.360.111"

8 Distributor

EFF